From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced)-2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

CRT – III

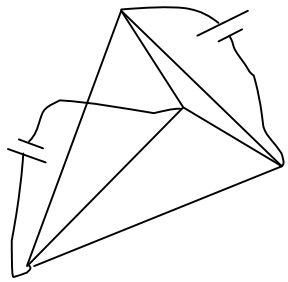
(Paper-1)

	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	A	C
2.	A	D	D
3.	B	D	B
4.	C	A	C
5.	A	B	C
6.	C	C	B
7.	B	B	A
8.	A, D	A, B, D	B, C
9.	A, B	A, D	A, C, D
10.	C, D	A, D	B, C
11.	B, C	A, B, C	B, C, D
12.	B	A	D
13.	B	C	C
14.	B	B	B
15.	D	B	A
16.	C	B	C
1.	3	5	4
2.	4	3	2
3.	5	4	9
4.	4	1	5
5.	1	3	7
6.	5	2	3
7.	2	3	5

Physics

PART – I

4. Apply superposition of currents in each resistor due to each cell to get current in two resistors zero, E/R in rest four, current drawn from each cell is 2E/R



5. One particle t/2 before max height, other t/2 after max height
Relative velocity perpendicular to line joining them = gt
Relative separation = ut
Relative angular velocity = gt/ut = g/u

6. Intensity in absence of lens = $P/4\pi r^2$
So intensity at P before placing lens is ¼ of intensity at optical center of lens
After crossing lens light becomes parallel rays to intensity does not change further along that ray

7. apply work-energy theorem

$$mgz - \left[\frac{1}{2}K\left(\frac{mg}{K}+z\right)^2 - \frac{1}{2}K\left(\frac{mg}{K}\right)^2\right] + Fz = 0$$

$\Rightarrow z = 2F/K.$

8. $\tau = MB \sin\theta$

$\Rightarrow \alpha \propto -\theta$

$$\Rightarrow f \propto \sqrt{\frac{i_1 i_2}{r_2}}$$

9. $a = kx^2$

$$\frac{da}{dx} = 2kx \Rightarrow \tan 60° = 2k\sqrt{3} \Rightarrow k = \frac{1}{2}$$

$$a = \frac{x^2}{2}$$

$$v\frac{dv}{dx} = \frac{x^2}{2} \Rightarrow \frac{v^2}{2} = \frac{x^3}{6} + C_1 \text{ at } x = 0,\ v = 3 \text{ m/s}$$

$$C_1 = \frac{9}{2}$$

$$v^2 = \frac{x^3}{3} + 9$$

hence a = 1.5 and $v = \sqrt{\sqrt{3}+9}$

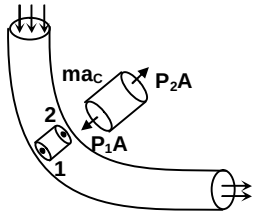
10. Capillary height $h = \dfrac{2T\cos\theta}{r\rho g}$

$\Rightarrow h = 59.6$ mm

here 59.6 mm is greater than the protruding part hence water will rise in the capillary of insufficient height 25 mm.

Now, $R = \dfrac{2T}{hrg} = 0.6$ mm

11. Fluid particles passing through the bend are in circular motion. The centripetal acceleration is provided by the variation in pressure. In the section as shown



$P_1A - P_2A = ma_C$

$\Rightarrow P_1 > P_2$

applying Bernoulli theorem at 1 and 2

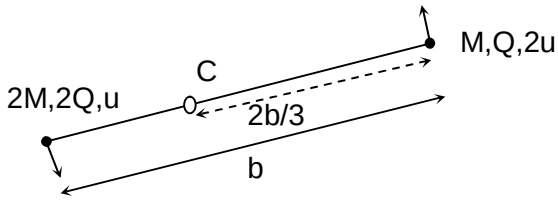
i.e., $P_1 + \dfrac{1}{2}\rho_1 v_1^2 + \rho g h_1 = P_2 + \dfrac{1}{2}\rho v_2^2 + \rho g h_2$

$P_1 + \dfrac{1}{2}\rho_1 v_1^2 = P_2 + \dfrac{1}{2}\rho v_2^2$ as $h_1 = h_2$

$\because \quad P_1 > P_2$

$v_1 < v_2$

12. In CM frame

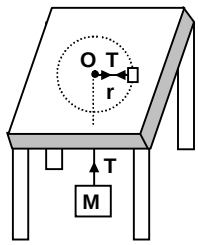


13. At closest approach velocity perpendicular to line joining particles
Conserve kinetic + electric potential energy from large separation to closest approach

16. System rotates about CM i.e. middle of 4M and M(–Q)
For problem 3, work done by electric field = QLE = ½ $I\omega^2$
Initial torque by electric field = QLE = $I\alpha$
$r\alpha/\omega^2 r = ½$

SECTION–C

1. For circular motion of a body tied to a string on a horizontal plane



$\dfrac{mv^2}{r} = T$

here, $T = mg$

$\dfrac{mv_1^2/r_1}{mv_2^2/r_2} = \dfrac{M_1 g}{M_2 g}$

$\dfrac{v_1^2 r_2}{v_2^2 r_1} = \dfrac{8}{1}$ …(i)

Now as angular momentum is conserved about pt 0.

$mv_1 r_1 = mv_2 r_2$ …(ii)

from (i) and (ii) $\dfrac{r_1}{r_2} = 2$

further $\frac{\omega_2}{\omega_1} = \frac{v_2}{v_1} \times \frac{r_1}{r_2} = \left(\frac{r_1}{r_2}\right)^2$

$\frac{\omega_2}{\omega_1} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$\frac{\Delta\omega}{\omega} = \frac{\omega_2 - \omega_1}{\omega_2} = \frac{\omega_2}{\omega_1} - 1 = \frac{1}{4} - 1 = -\frac{3}{4}$

2. $\frac{\Delta Q}{\Delta t} = K_A A \frac{dT}{dx}$

$\frac{dT}{dx}$ in conductor A = slope of graph = $\sqrt{3}$

Since both conductors are connected in series, same heat current will flow in A and B.

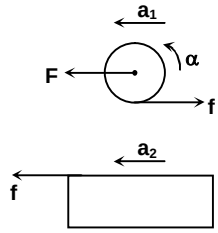
$K_B = \frac{\Delta Q}{\Delta t} \times \frac{1}{A\frac{dT}{dn}} = \frac{\Delta\theta}{\Delta t} \times \frac{1}{A\tan 45°} = 200\sqrt{3}$ J/m-sec°C

3. For lens L_1, ray must move parallel to the axis after refraction $\frac{\mu_1}{\infty} + \frac{\mu_w}{x} = \frac{\mu_1 - \mu_\omega}{R_1} \Rightarrow x = 10$ cm

4. $f = ma_2$

$\alpha = \frac{\tau}{I} = \frac{2f}{mR}$

$\Rightarrow$ $a_2 = F/4m$, $f = F/4$



5. In steady state potential difference across each capacitor = E

6. $m_{Ag}\sin\theta = 5 \times 10 \times \sin 37° = 5 \times 10 \times \frac{3}{5} = 30$ N

$f_{max} = \mu_{AB}(m_C + m_B)g\cos\theta = 48$ N

hence f = 30 N

7. $|\sin\theta| \le 1$

$-1 \le 2\left[\frac{n}{40} - 1\right] \le 1$

or $20 \le n \le 60$

Hence number of maxima = 60 – 20 = 40

Chemistry

PART – II

1. $\Delta G^o = -RT\ln K_p$

$\Rightarrow 46.5 \times 1000 = -2 \times 298 \times \ln K_p$

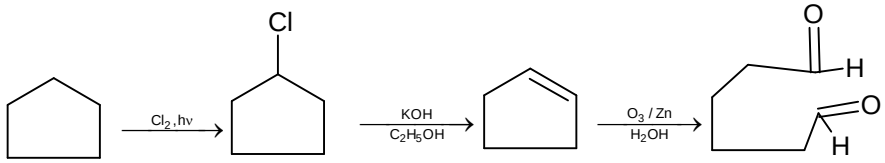
$\Rightarrow K_p = 1.3 \times 10^{-34}$

2. $C_6H_6 + \frac{15}{2}O_2 \longrightarrow 6CO_2 + 3H_2O; Q_1$

$C + O_2 \longrightarrow CO_2; \quad Q_2$

$H_2 + \frac{1}{2}O_2 \longrightarrow H_2O; \quad Q_3$

Required equation: $6C(s) + 3H_2 \longrightarrow C_6H_6;\ \Delta H = 6Q_2 + 3Q_3 - Q_1$

3. 

4. Mole fraction of A in vapour $Y_A = \frac{X_A P_A^o}{X_A P_A^o + X_B P_B^o}$

$\because \frac{X_A}{X_B} = \frac{1}{3} \& \frac{P_A^o}{P_B^o} = \frac{1}{3}$

$\Rightarrow Y_A = \frac{1}{1 + \frac{X_B}{X_A}\frac{P_B^o}{P_A^o}} = \frac{1}{1+3.3} = 0.1$

6. $[Cu(CN)_4]^{3-}$ is square planar and contains Cu (I) which is colourless
$[NiCl_4]^{2-}$ is tetrahedral and contains Ni(II) which is coloured.
$[BF_4]^-$ is tetrahedral and colourless

14. $SO_2 + PCl_5 \longrightarrow \underset{(A)}{SOCl_2} + POCl_3$

$SOCl_2 + H_2O \longrightarrow SO_2 + HCl$

(A) (pungent smelling)

15. $M_{SOCl_2} = \frac{5.95}{119 \times 1} = 5 \times 10^{-2}\ mol/l$

$H_2O + SOCl_2 + 2AgNO_3 \longrightarrow 2AgCl + 2HNO_3 + SO_2$

Moles of AgCl formed = 2 × molarity of $SOCl_2$ in 100 ml = $2 \times 5 \times 10^{-2} \times \frac{100}{1000} = 10^{-2}$ moles

Wt. of AgCl = $143.5 \times 10^{-2} = 1.435$ g

16. IInd part was treated with dil. $HCl/BaCl_2/H_2O_2$.

$SOCl_2 + H_2O \longrightarrow SO_2 + 2HCl$

$H_2O_2 + SO_2 \longrightarrow SO_3 + H_2O$

$SO_3 + H_2O \longrightarrow H_2SO_4$

$BaCl_2 + H_2SO_4 \longrightarrow BaSO_4 + 2HCl$

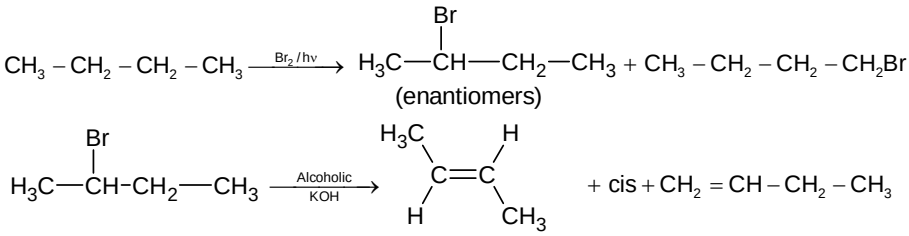
ppt.

Mole of $BaSO_4$ formed = $5\times10^{-2}\times\frac{10}{100}$ in 100 ml solution

$\therefore$ Wt. of $BaSO_4$ formed = $5\times10^{-3}\times233 = 1.165$ g.

SECTION – C

2.



3. Let x% be the decrease in density of Ge crystals. Let 'a' be the total no. of Ge atoms missing and 'b' be the no. of boron atoms replacing Ge atoms.

$\frac{b}{a}\times100 = 2.376$ … (i)

$1-\frac{72.6a}{N_{AV}}+\left(\frac{11b}{N_{AV}}\right)=1-0.01x$

$\frac{72.6a}{N_{AV}}-\frac{11b}{N_{AV}}=0.01x$ … (ii)

Also, $b = \frac{150N_{AV}}{10^6\times11}\times(1-0.01x)$ … (iii)

$\therefore$ Solving (i), (ii) and (iii)

x = 4%

4. $K_{sp} = [Zn^{2+}][S^{2-}]$

$[S^{2-}] = \frac{10^{-21}}{0.01} = 10^{-19}$

For $K_{a_1}.K_{a_2} = \frac{\left[H^+\right]^2\left[S^{2-}\right]}{\left[H_2S\right]}$

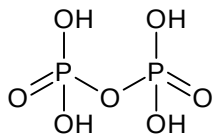
$10^{-20} = \frac{\left[H^+\right]^2\times10^{-19}}{0.1} \Rightarrow \left[H^+\right] = 0.1$

or pH = 1

5. $3Cu + 10HNO_3 \rightarrow 3Cu(NO_3)_2 + 3NO_2 + NO + 5H_2O$

6. Only CO and CH_4 react with O_2, N_2 remain uncreative

7.



$\frac{6}{2} = 3$

Mathematics PART – III

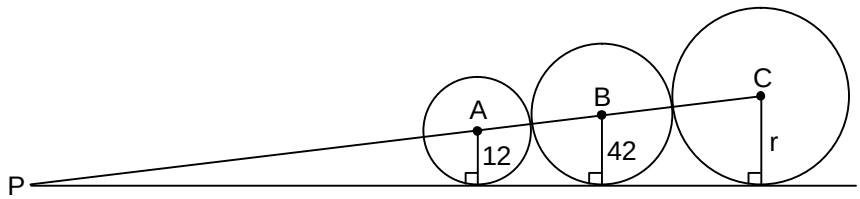
1. A, B, D are always true but C is true only if the triangle is equilateral

2. $f(x) = \cos x + \cos(x + 2\alpha) + \cos(x + \alpha) = (2\cos\alpha + 1)\cos(x + \alpha)$
As $\cos(x + \alpha)$ can take any real value from –1 to 1 $\forall\ x \in R$
But f(x) is constant so $(2\cos\alpha + 1) = 0$ must hold
$\Rightarrow \alpha = \frac{2\pi}{3}$ and C = 0
Hence $[C + \alpha] = \left[0 + \frac{2\pi}{3}\right] = 2$

3. $\int \frac{2\sin 3x\cos x - 2\sin 3x\cos 3x}{2\cos^2 x + 2\cos x\cdot\cos 5x}dx$
$\int \frac{\sin 3x(\cos x - \cos 3x)}{\cos x(\cos x + \cos 5x)}dx$
$\int \frac{2\sin 3x\cdot\sin 2x\cdot\sin x}{2\cos x\cdot\cos 3x\cdot\cos 2x}dx$
$\int \tan 3x\cdot\tan 2x\cdot\tan x\ dx = \int(\tan 3x - \tan 2x - \tan x)dx$

4. $\beta(2) = \int_0^{2\pi}\sqrt{1-\sin t}dt$
$\beta(2) = \int_0^{\pi}\sqrt{1-\sin t}dt + \int_{\pi}^{2\pi}\sqrt{1-\sin t}dt$
$\beta(2) - \beta(1) = \int_{\pi}^{2\pi}\sqrt{1-\sin t}dt$
Put $t = \pi + x$, $\beta(2) - \beta(1) = \int_0^{\pi}\sqrt{1+\sin x}dx$
$= \int_0^{\pi}\left|\cos\frac{x}{2} + \sin\frac{x}{2}\right|dx = 2\left[\sin\frac{x}{2} - \cos\frac{x}{2}\right]_0^{\pi} = 4$

5. We have $\sin\theta = \frac{12}{PA} = \frac{42}{PB} = \frac{r}{PC} = K(\text{Let})$



So K(PB – PA) = 42 – 12 = 30
K(AB) = 30
K[54] = 30
$K = \frac{30}{54} = \frac{5}{9}$

Now K(PC – PB) = r – 42
K(BC) = r – 42
K[r + 42] = r – 42
On solving r = 147

6. Let $g(t) = \int_0^t f(x)dx$

$g'(t) = f(t)$
Now apply Lagrange's mean value theorem in [0, 2], we get
$\frac{g(2)-g(0)}{2-0} = g'(c)$ where $c \in (0, 2)$
$\Rightarrow \int_0^2 f(x)dx = 2g'(c)$ where $c \in (0, 2)$

7. Vector perpendicular to $(2\hat{i}+\hat{j}-\hat{k})$ and $(\hat{i}+3\hat{j}-\hat{k})$ is

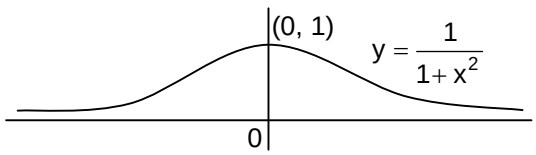
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 3 & -1 \end{vmatrix} = 2\hat{i}+\hat{j}+5\hat{k}$$

Any general point on the line is $(1 + 2\lambda, 1 + \lambda, 1 - \lambda)$ at the point of intersection. This point satisfy equation of plane $(1 + 2\lambda) + 3(1 + \lambda) - (1 - \lambda) = 9$
$\Rightarrow \lambda = 1$
Point of intersection (3, 2, 0) and required line $\frac{x-3}{2} = \frac{y-2}{1} = \frac{z}{5}$

8. (A) $f(x) = \cos(\sin x)$
As $f(x + \pi) = \cos(\sin(\pi + x)) = \cos(-\sin x) = \cos(\sin x) = f(x)$
(B) As $-1 \le \sin x \le 1$
$\cos 1 \le \cos(\sin x) \le 1$
(C) $\lim_{x\to\pi/2}\left[f\left(\frac{\pi}{2}-x\right)+f\left(\frac{\pi}{2}+x\right)\right]$
$= \lim_{x\to\pi/2} 2\cos(\cos x) = 2$
(D) Clearly f(x) is even function

9. Let the number be (a – d), a, (a + d)
3a = 24
a = 8
Also (8 – d)(8)(8 + d) = 384
$d^2 = 16$
d = 4 or –4
Hence series is 4, 8, 12, ...
$\Rightarrow S_n = 2n(n + 1)$ or 12, 8, 4, ...
$\Rightarrow S_n = 14n - 2n^2$

10. $f(x) > 0$ and $f'(x) > 0$ for $x < 0$
$f'(x) < 0$ for $x > 0$
Now $g(x) = \frac{1}{f\left(\frac{1}{x}\right)}$

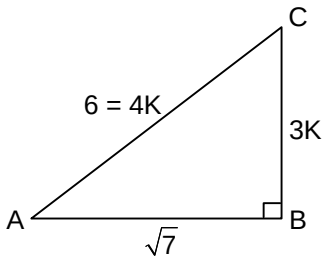


$$g'(x) = \frac{-1}{f^2\left(\frac{1}{x}\right)} \times f'\left(\frac{1}{x}\right) \times \left(\frac{-1}{x^2}\right)$$

So for $x < 0 \Rightarrow g'(x) > 0$

$x > 0 \Rightarrow g'(x) < 0$

11. As ac = 3 and bc = 4



$$\frac{b}{a} = \frac{4}{3} \Rightarrow \frac{b-a}{b+a} = \frac{4-3}{4+3} = \frac{1}{7} \quad (\angle B > \angle A)$$

$$\tan\left(\frac{B-A}{2}\right) = \frac{b-a}{b+a}\cot\frac{C}{2}$$

$$\sqrt{\frac{1-\cos(B-A)}{1+\cos(B-A)}} = \frac{4-3}{4+3}\cot\frac{C}{2}$$

On solving $\cot\frac{C}{2} = \sqrt{7} \Rightarrow \cos C = \frac{3}{4}$

$\therefore$ cos C = cos(A – B) or cos C = cos(B – A)

C = A – B C = B – A

$\pi - (A + B) = A - B$ $\pi - (A + B) = B - A$

$A = \frac{\pi}{2}$ (rejected) $B = \frac{\pi}{2}$

But $\frac{b}{a} = \frac{4}{3} \Rightarrow \frac{b}{4} = \frac{a}{3} = K$

Now $ac = 3 \Rightarrow 3\sqrt{7}K^2 = 3 \Rightarrow K = \frac{1}{7^{1/4}}$

$$R = 2K = \frac{2}{7^{1/4}}$$

12. $y = f(x) = (\ln x)^2 + 2 \ln x$

$$\left.\frac{dy}{dx}\right|_{A(a,\, f(a))} = \frac{2\ln a}{a} + \frac{2}{a} = \frac{2}{a}(1 + \ln a)$$

Also $\frac{dy}{dx} = \frac{f(a) - 0}{a - 0}$

$\frac{f(a)}{a} = \frac{2}{a}(1 + \ln a) \Rightarrow f(a) = 2(1 + \ln a)$

So $(\ln a)^2 + 2 \ln a = 2 + 2 \ln a$

$(\ln a)^2 = 2 \Rightarrow a = e^{\sqrt{2}}$ or $e^{-\sqrt{2}}$

Hence $a = e^{-\sqrt{2}}$ and $b = e^{\sqrt{2}}$

So ab = 1

13. $5x \cdot f'(x) - x \cdot \ln 10 - 10 = 0$

$$5x\left[\frac{2\ln x}{x} + \frac{2}{x}\right] - x\ln 10 - 10 = 0$$

$10 \ln x + 10 - x \ln 10 - 10 = 0$

$10 \ln x = x \ln 10$

$\frac{\ln x}{x} = \frac{\ln 10}{10}$, x > 0

As $y = \frac{\ln x}{x}$

$\Rightarrow y' = \frac{1-\ln x}{x^2}$ so two values of x are possible

14.-16 We have $f(x) = \lim_{\alpha\to 0} \frac{e^x}{\alpha^4} \int_0^{\alpha} \frac{(e^t-1)\ln^2(1+t)}{(2t^3+3)}\,dt$ $\left[\frac{0}{0} \text{ form}\right]$

$= \lim_{\alpha\to 0} \frac{e^x(e^\alpha-1)\ln^2(1+\alpha)}{(2\alpha^3+3)4\alpha^3}$

$= e^x \lim_{\alpha\to 0} \frac{e^\alpha-1}{\alpha}\cdot\frac{\ln^2(1+\alpha)}{\alpha^2}\cdot\frac{1}{4(2\alpha^3+3)}$

$= \frac{e^x}{12}$

14. Clearly f(ln 2) = $\frac{e^{\ln 2}}{12} = \frac{1}{6}$

15. We have $\int_0^x g(t)\,dt = 3x + \int_x^0 \cos^2 t\cdot g(t)\,dt$

On differentiating both side w.r.t x

$g(x) = 3 - \cos^2 x\, g(x)$

$g(x) = \frac{3}{1+\cos^2 x}$

Clearly $\frac{3}{2} \le \frac{3}{1+\cos^2 x} \le 3$

16. Let $I = \int_0^{\pi/2} g(x)\,dx$

$= \int_0^{\pi/2} \frac{3}{1+\cos^2 x}\,dx$

$= \int_0^{\pi/2} \frac{3\sec^2 x}{\tan^2 x+2}\,dx$

$= \frac{3\pi}{2\sqrt{2}}$

SECTION – C

1. $T_k = \tan^{-1}\left(\frac{8}{k^2}\right) = \tan^{-1}\left(\frac{\frac{2}{k^2}}{4}\right) = \tan^{-1}\left[\frac{2}{1+\left(\frac{k^2}{4}-1\right)}\right]$

$= \tan^{-1}\left[\frac{\left(\frac{k}{2}+1\right)-\left(\frac{k}{2}-1\right)}{1+\left(\frac{k}{2}+1\right)\left(\frac{k}{2}-1\right)}\right]$

$= \tan^{-1}\left[\frac{k}{2}+1\right] - \tan^{-1}\left[\frac{k}{2}-1\right]$

2. $\frac{a}{\cos(200^o)} - \frac{c\sin(200^o)}{\cos(200^o)} = d$ ..... (1)

$\frac{b}{\cos(200^o)} + \frac{d\sin(200^o)}{\cos(200^o)} = c$ ..... (2)

$a = d\cos(200^o) + c\sin(200^o)$ ..... (3)

$b = c\cos(200^o) - d\sin(200^o)$ ..... (4)

$\therefore a^2 + b^2 = c^2 + d^2$

$\therefore (a^2 + b^2 + c^2 + d^2) = 2(c^2 + d^2)$

$ac = dc\cos(200^o) + c^2\sin(200^o)$

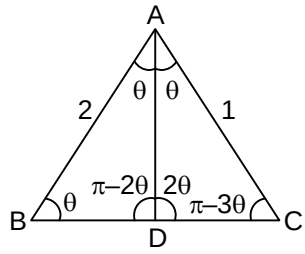
Also $bd = cd\cos(200^o) - d^2\sin(200^o)$

$ac - bd = (c^2 + d^2)\sin(200^o)$

So $\frac{a^2 + b^2 + c^2 + d^2}{ac - bd} = \frac{2(c^2 + d^2)}{(c^2 + d^2)\sin(200^o)} = 2\,\text{cosec}(200^o)$

$\therefore p = 2$

3. Since $\frac{AB}{AC} = \frac{BD}{DC} = \frac{2}{1}$



Using since law in ΔABC we get

$\frac{1}{\sin\theta} = \frac{2}{\sin(\pi - 3\theta)} = \frac{2}{\sin 3\theta}$

$3 - 4\sin^2\theta = 2$

$\Rightarrow \sin^2\theta = \frac{1}{4} \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30^o \Rightarrow 2\theta = 60^o$

Area of $\Delta ABC = \frac{1}{2} \times 2 \times 1 \times \sin 60^o = \frac{\sqrt{3}}{2}$

Hence $12\Delta^2 = 12 \times \frac{3}{4} = 9$

4. Since diagonal is a member of both the families so it will pass through (1, 0) and (–1, 1)
Equation of diagonal AC is $x + 2y - 1 = 0$
Since one of the vertex (3, 2) which does not be on AC, so equation of BD is $2x - y = 4$

Point of intersection of AC and BD is $P\left(\frac{9}{5}, -\frac{2}{5}\right)$

If vertex B is (3, 2) then vertex D is $\left(\frac{3}{5}, -\frac{14}{5}\right)$ also $BD = \frac{12\sqrt{5}}{5}$ (say d_1)

Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$

$\frac{1}{2} \times \left(\frac{12\sqrt{5}}{5}\right) \times d_2 = 12\sqrt{5}$

$d_2 = 10$

5. We have $f'(x) = f'(3 - x)$ on integrating w.r.t. x
$f(x) = -f(3 - x) + c$
Put $x = 0$
$f(0) + f(3) = c$
$-32 + 46 = c$
$c = 14$

Now let $I = \int_0^3 f(x)\,dx$ ..... (1)

$I = \int_0^3 f(3-x)\,dx$ ..... (2)

Adding (1) and (2), we get $2I = \int_0^3 (f(x) + f(3-x))\,dx = \int_0^3 14\,dx = 42$

6. $\frac{a^3+b^3}{ab^2} = \frac{\sin^3 20^o + \sin^3 80^o}{\sin 20^o \sin^2 80^o}$

$= \frac{3\sin 20^o - \sin 60^o + 3\sin 80^o - \sin 240^o}{4\sin 20^o \cos^2 10^o}$

$= \frac{3(\sin 20^o + \sin 80^o)}{4\sin 20^o \sin^2 80^o} = \frac{3\cos 30^o}{4\sin 40^o \sin 20^o \sin 80^o}$

$= \frac{3}{4}\frac{\cos 30^o}{\sin 20^o \sin(60^o - 20^o)\,s\sin(60^o + 20^o)}$

$= \frac{3}{4}\frac{\cos 30^o}{\frac{1}{4}\sin 60^o} = 3$

7. $f'(x) = 6P - 4P\cos 4x - 5 - 3\cos 3x > 0$

$2P > \frac{5 + 3\cos 3x}{3 - 2\cos 4x}$

2P > max value of $\frac{5 + 3\cos 3x}{3 - 2\cos 4x}$

$2P > \frac{5+3}{3-2}$

P > 4

Smallest P is 5